

```
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ./practical9
```

```
=====
PRACTICAL 9
=====
```

Commands:

```
copy      - Low-level vs standard file copy
redirect  - Demonstrate dup2() redirection
all       - Execute both parts
exit      - Exit program
```

```
practical9> copy
```

```
=====
PART A - FILE COPY TEST
=====
```

```
Source file created: practical9_input.txt
lseek() final source offset: 358894 bytes
```

Low-level copy: SUCCESS

System calls: open() read() write() lseek() close()

Low-level copy time: 0.000771 seconds

Standard library copy: SUCCESS

Functions: fopen() fread() fwrite() fclose()

Standard library copy time: 0.000812 seconds

```
===== COPY VERIFICATION =====
```

Original file : 358894 bytes

Low-level copy: 358894 bytes

Standard copy : 358894 bytes

Copy verification: PASS

```
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```

```
practical9> redirect
```

```
=====
PART B - I/O REDIRECTION
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```
Input/output redirection completed.
```

```
STDIN -> redirect_input.txt
```

```
STDOUT -> redirect_output.txt
```

```
dup2() redirection: PASS
```

```
===== OUTPUT FILE CONTENT =====
```

```
===== REDIRECTED OUTPUT =====
```

```
READ: Hello from redirected input.
```

```
READ: This data is being read through STDIN.
```

```
READ: This demonstrates how a shell performs input redirection.
```

```
=====
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```

```
practical9>
```

```
Practical 9 completed successfully.
```

```
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ cd 2520090020_Practical
-bash: cd: 2520090020_Practical: No such file or directory
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ cd oslab
-bash: cd: oslab: No such file or directory
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ cd 2520090020_Practical
-bash: cd: 2520090020_Practical: No such file or directory
tarun@LAPTOP-BRD2JBFC:/mnt/c/WINDOWS/system32$ cd ~/oslab/2520090020_Practical
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ls -l practical9_*
-rw-r--r-- 1 tarun tarun 358894 Oct  7 03:05 practical9_input.txt
-rw-r--r-- 1 tarun tarun 358894 Oct  7 03:05 practical9_low.txt
-rw-r--r-- 1 tarun tarun 358894 Oct  7 03:05 practical9_std.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ls -l redirect_*
-rw-r--r-- 1 tarun tarun 126 Oct  7 03:05 redirect_input.txt
-rw-r--r-- 1 tarun tarun 225 Oct  7 03:05 redirect_output.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ cat redirect_output.txt
===== REDIRECTED OUTPUT =====
READ: Hello from redirected input.
READ: This data is being read through STDIN.
READ: This demonstrates how a shell performs input redirection.
=====
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$
```